

SmartChef
Your Personal Cooking Companion

From "What can I make?" to plating — powered by intelligent ingredient scanning

Project Brief & Objectives



Design a Seamless Experience

Create a mobile app that helps home cooks effortlessly decide what to cook, eliminating the daily kitchen dilemma through intelligent technology.



Reduce Friction

Leverage ingredient scanning and personalized smart suggestions to minimize decision fatigue and streamline the cooking process from start to finish.



Support End-to-End Flow

Build intuitive interactions that guide users through every stage of cooking, from ingredient discovery to final plating.



Deliver Core Features

High-fidelity designs for Ingredient Scan and Recipe Library, plus wireframes for Timers and Grocery List functionality.

The Problem

Home cooks face a frustrating daily challenge that impacts their relationship with cooking. The question "What should I make for dinner?" creates stress and wastes valuable time.

- People struggle daily with deciding what to cook based on available ingredients
- Searching for recipes online is time-consuming and overwhelming
- Managing multiple steps, timers, and ingredients breaks the natural cooking flow
- This results in wasted food, wasted time, and reduced motivation to cook at home

Without a streamlined solution, the joy of cooking gets lost in the friction of planning and execution.



What We Learned About Home Cooks

Through user research and interviews, we uncovered key insights that shaped our design approach. Understanding the real needs and behaviors of home cooks was essential to creating a solution that truly resonates.

Speed Without Compromise

Home cooks want quick meal solutions but refuse to sacrifice variety or quality. They're looking for efficiency that doesn't limit their culinary creativity.

Minimal Friction to Start

The easier it is to begin cooking, the more likely they are to do it. Every extra step or decision point becomes a barrier to actually making a meal.

Cook With What's Available

Most people prefer to cook with ingredients they already have at home rather than making special trips to the store for specific recipes.

Visual Learning Style

When following recipes, users rely heavily on visual cues and imagery. Clear photos and visual hierarchy are more important than lengthy text instructions.

Meet Rohan — Our Primary Persona



Rohan, 28

Marketing Manager

Location: Mumbai, CA

Tech-savvy: Early adopter of smart home devices

Background & Lifestyle

Rohan is a busy working professional juggling a demanding career with personal wellness goals. He values his time and wants to maintain a healthy lifestyle without spending hours in the kitchen.

Pain Points

- Limited awareness of what's actually in his pantry and fridge
- Wastes time searching for recipes that match his ingredients
- Often orders takeout due to decision fatigue
- Wants to reduce food waste and eat healthier

Goals & Motivations

- Prepare fast, nutritious meals without extensive planning
- Make better use of groceries before they expire
- Find recipes that fit his dietary preferences
- Streamline the cooking process from decision to plate

Our Solution — SmartChef

SmartChef is your AI-powered cooking companion that transforms how you approach meal preparation. By combining intelligent ingredient recognition with personalized recipe matching, we've created an end-to-end solution that removes friction from every stage of cooking.



Discover Meals Instantly

Point your camera at ingredients and let AI instantly identify them. Get personalized recipe suggestions based on what you actually have at home.



Manage Timers Effortlessly

Auto-generated timers sync with recipe steps. View all active timers in one dashboard, perfect for managing multiple dishes simultaneously.



Cook Step-by-Step

Follow clear, sequential guidance designed for the kitchen environment. Large text, voice-friendly navigation, and intuitive progress tracking keep you on track.



Auto-Generated Lists

Convert any recipe into a categorized grocery list instantly. Edit quantities, check off items, and share with family members seamlessly.

The End-to-End Journey

We mapped the complete user experience from opening the app to completing a meal. This journey visualization helped us identify opportunities to reduce friction and create a seamless flow.



Open App

Quick access to scan feature on home screen

Scan Ingredients

Live camera detection with real-time labeling



Review & Edit

Confirm detected items, manually add missing ingredients

Browse Recipes

View personalized suggestions ranked by ingredient match



Start Cooking

Enter guided cooking mode with step-by-step instructions

Set Timers

Auto-generated timers linked to recipe steps

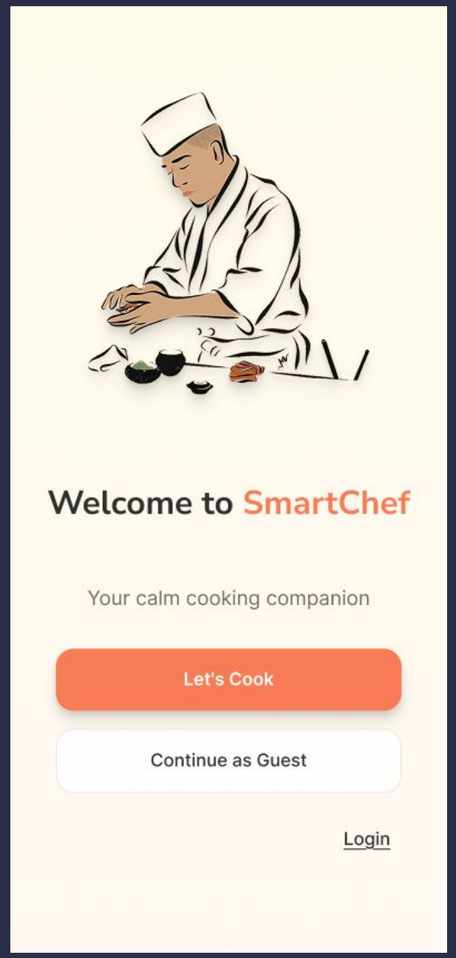
Add to List

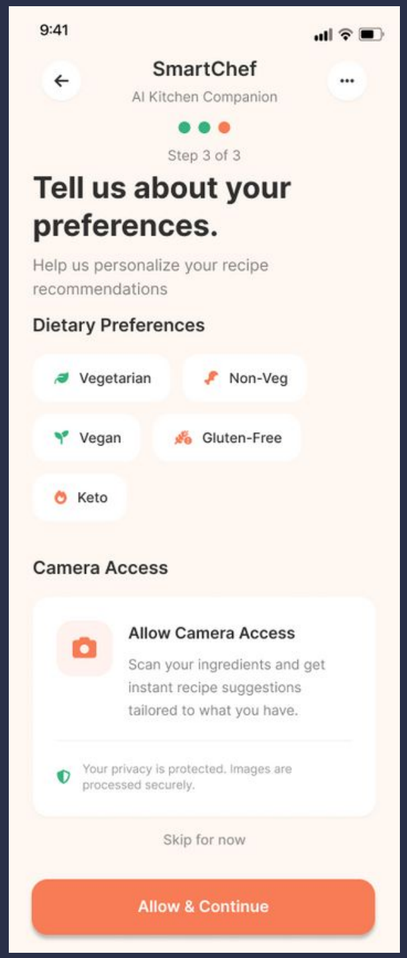
Missing ingredients automatically added to grocery list

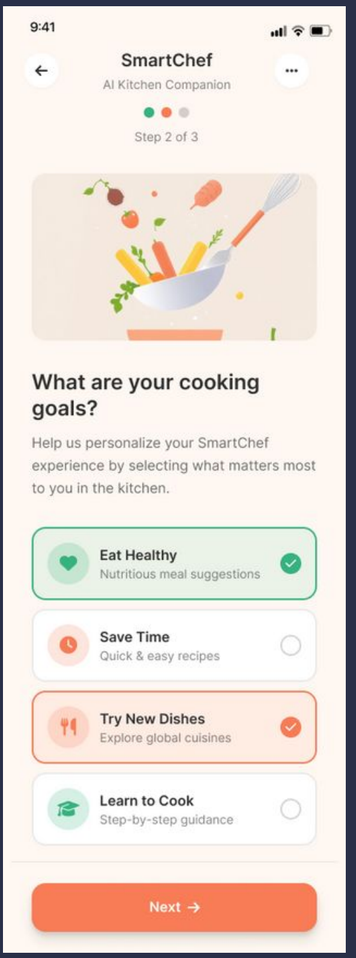
Onboarding

The onboarding flow is intentionally kept to 3 lightweight steps to reduce friction and get users into the product quickly. Each screen captures one high-signal input without overwhelming the user.









Home Screen

The home screen is designed to reduce decision fatigue and guide users toward the fastest path to cooking. Every element is ordered by frequency of use and expected user intent.

Personalized Greeting

A friendly “Hi, Rohan 👋” creates familiarity and reinforces this as a personal cooking companion, not a generic recipe app. It sets a welcoming, low-pressure tone.

Primary Action — Scan Ingredients

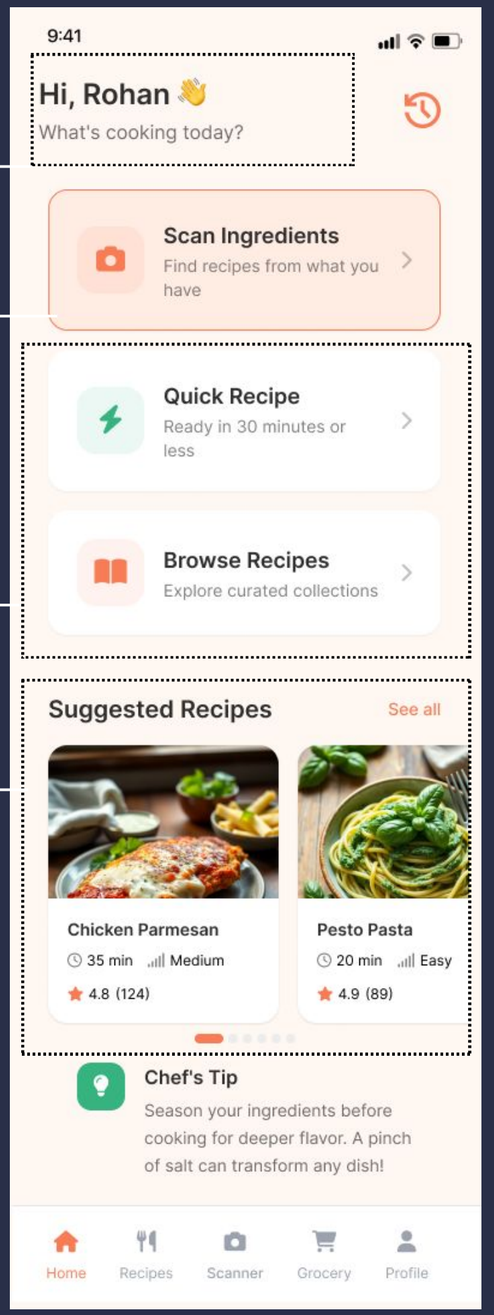
Positioned at the top because scanning is the highest-value feature and the fastest route to relevant recipes. The large card, warm color, and icon ensure it stands out as the default action.

Quick Recipe & Browse Recipes

These secondary actions are intentionally smaller and neutral-toned to avoid competing with scanning. They represent alternate user goals (speed or exploration) without overwhelming the hierarchy.

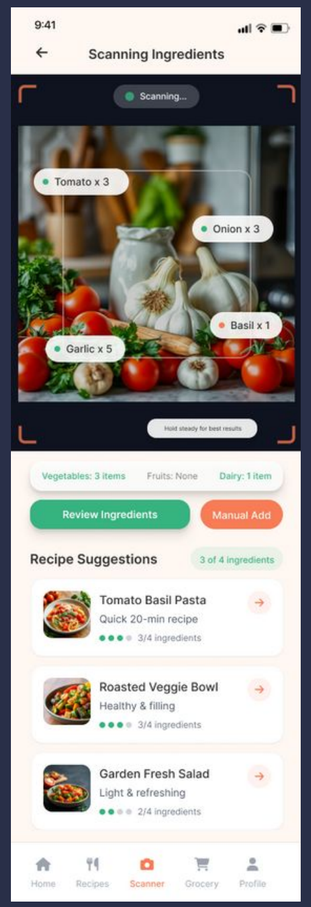
Suggested Recipes Carousel

Surfaces top recommendations based on user behavior, pantry history, or goals. Recipe cards highlight key decision factors—time, difficulty, and rating—to support rapid scanning and comparison.

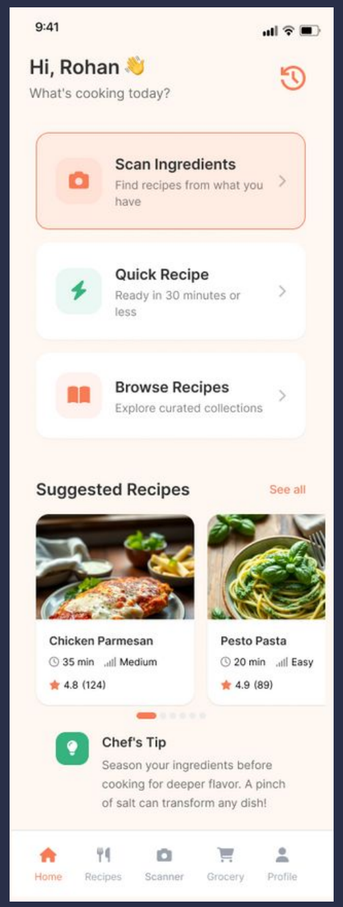


Feature One: Ingredient Scan & Smart Suggestions

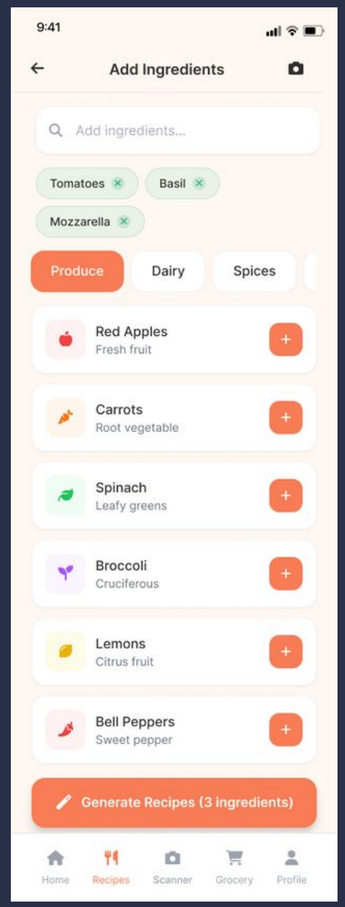
The ingredient scanning flow is designed to minimize friction and reduce the time between “What do I cook?” and “Here are your best recipe options.” Every part of the flow prioritizes speed, clarity, and real-world kitchen usability.



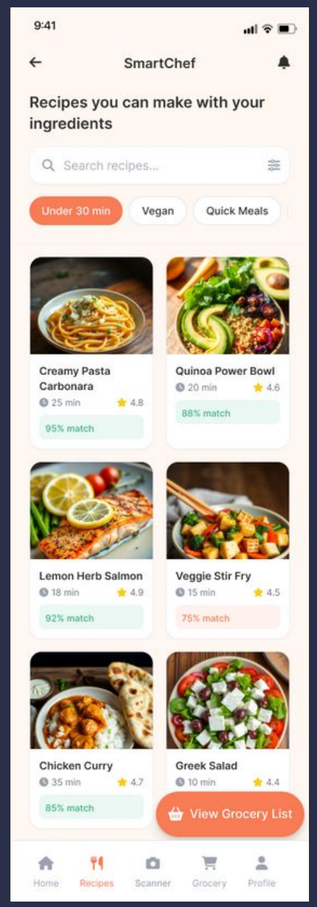
Live Ingredient Detection



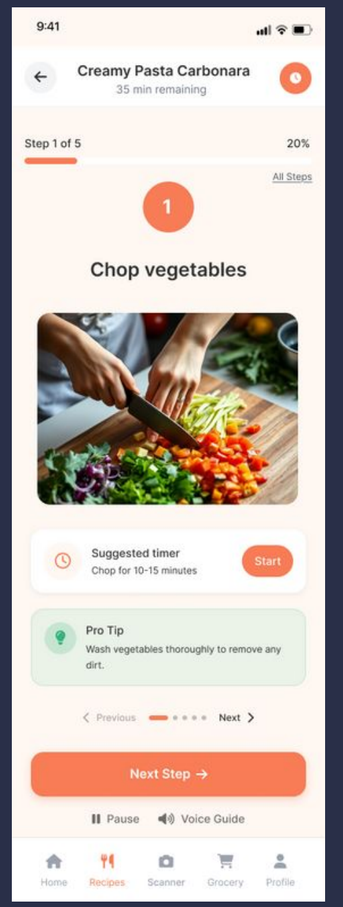
Manual Add as a Fail-Safe



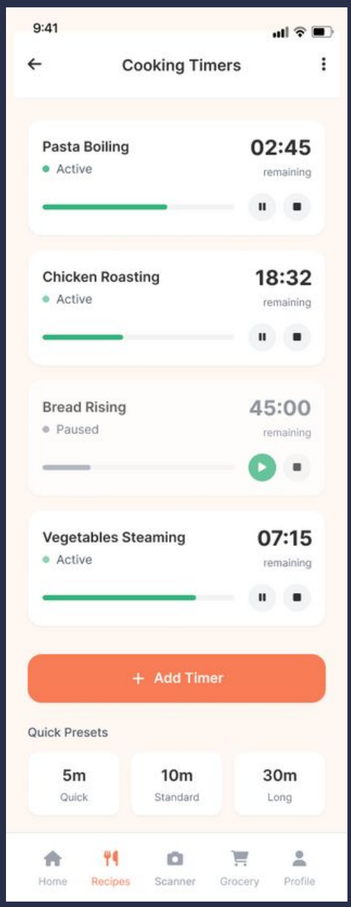
Ingredient Summary



Smart Recipe Suggestions



Recipe Generation



Timer based Steps

Ingredient Scan & Smart Suggestions

Focused, Camera-First Layout

The live camera occupies the majority of the screen so users can position ingredients without distraction. Minimal UI around it keeps attention on capturing the items clearly

Real-Time Ingredient Tags

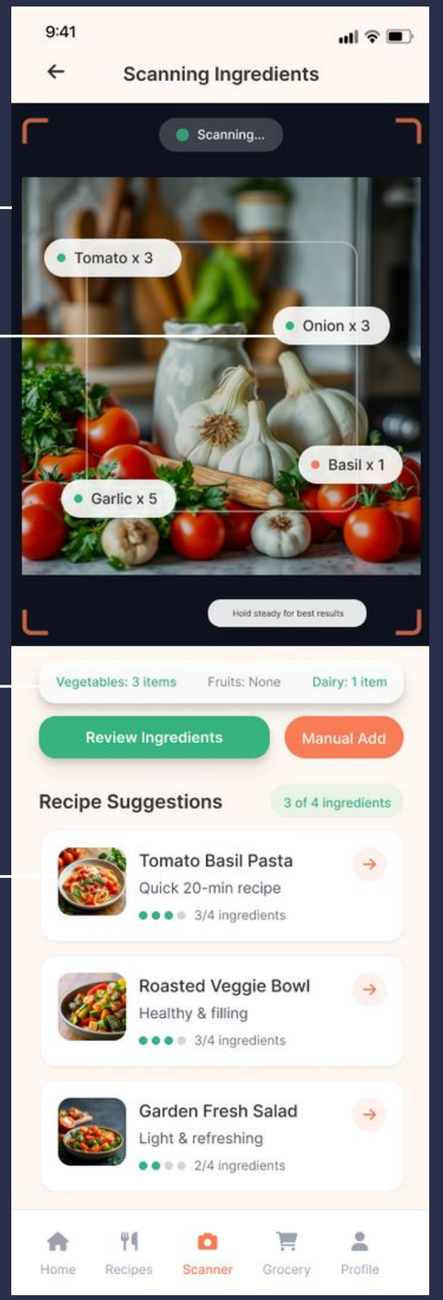
Overlay labels like Tomato × 3 and Garlic × 5 confirm detections instantly. This immediate feedback builds trust in the scanner and reduces the need for manual corrections

Category Summary Chips

Chips like Vegetables: 3 items help users understand what's detected at a glance. This mirrors how people mentally group groceries and reassures them that nothing important was missed

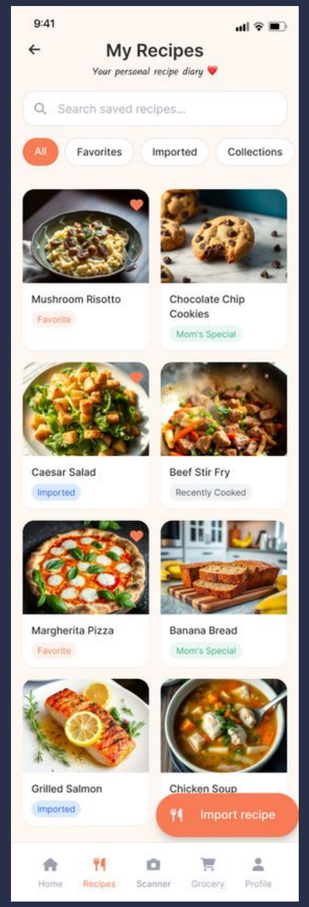
Instant Recipe Suggestions

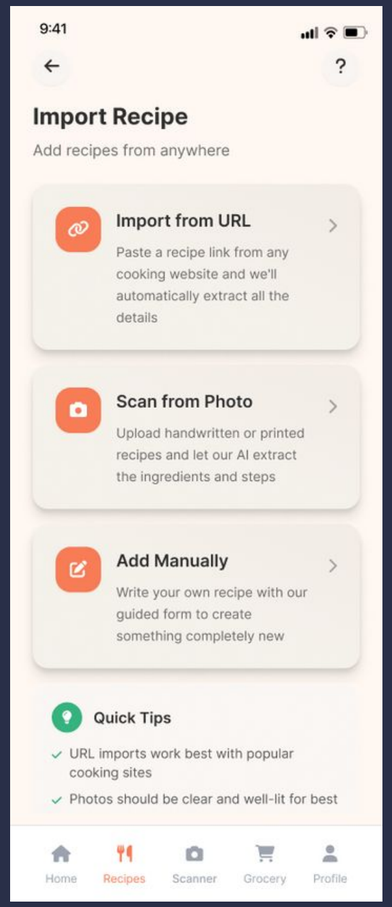
Showing recipes even before the user reviews ingredients shortens the path from scanning → action. Match indicators (3/4 ingredients) make the AI logic transparent and reduce hesitation



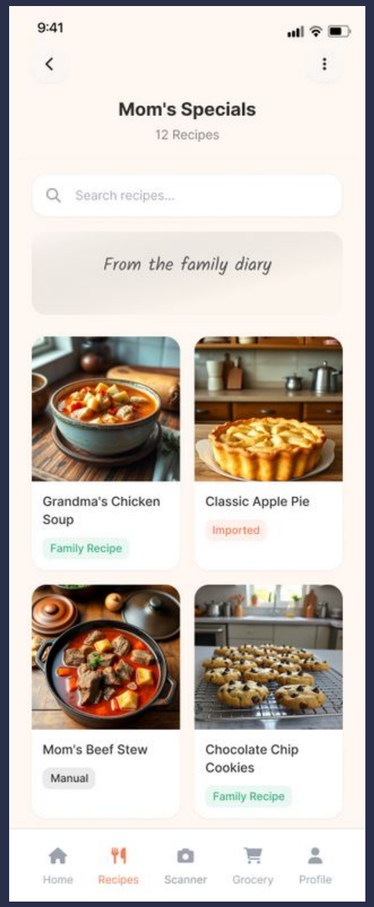
Feature Two: Recipe Library

The Recipe Library is designed to feel like a modern, searchable version of a family recipe diary—personal, emotional, and easy to browse. The goal is to help users store, organize, and rediscover recipes with minimal friction, while keeping the experience warm and nostalgic.

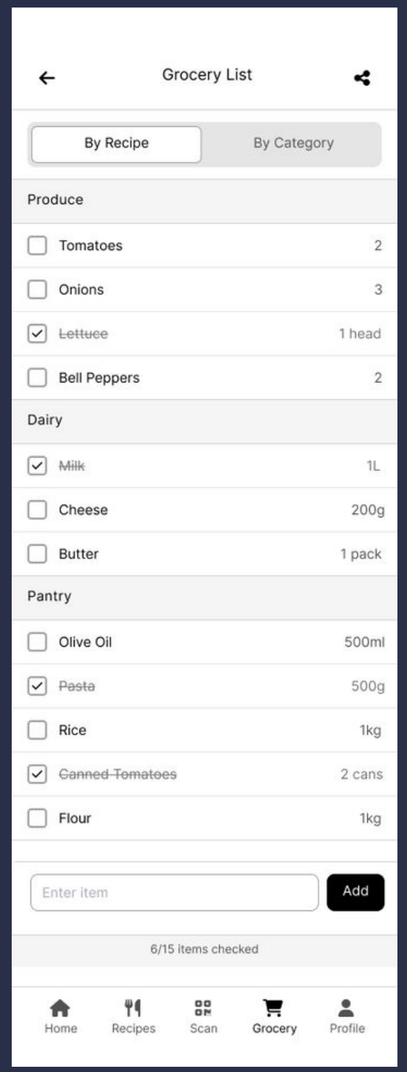


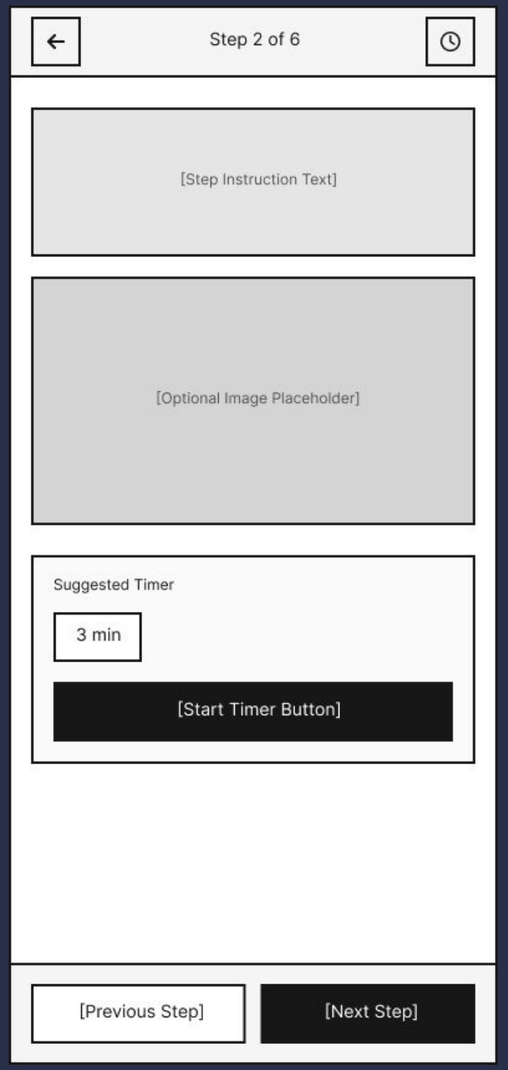


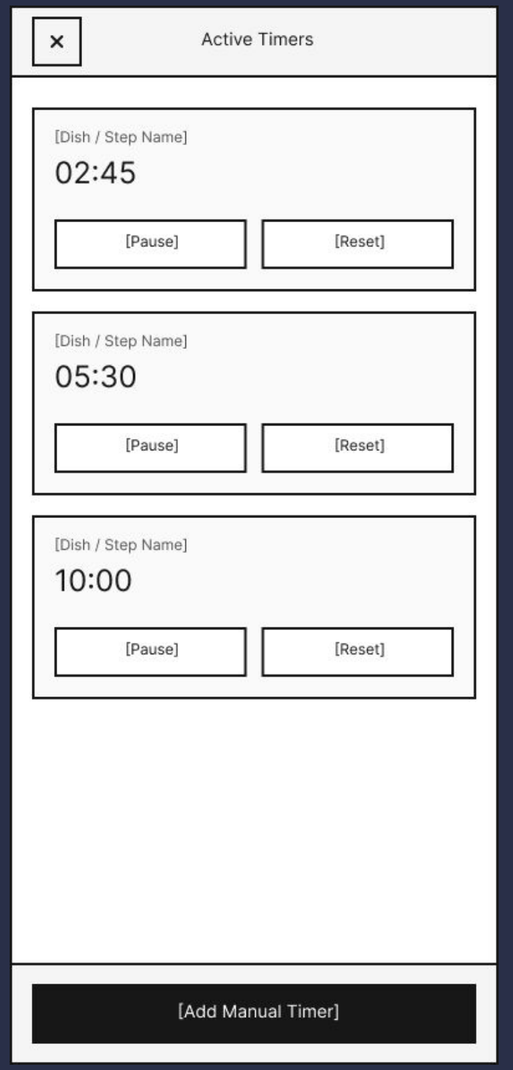


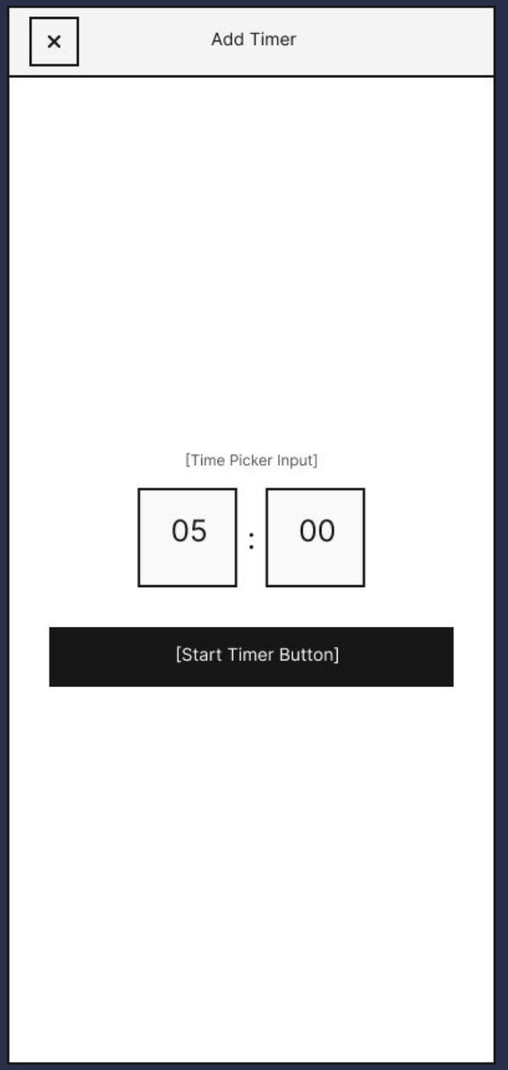


Wireframe (Grocery List + Timers).









Grocery List

Timers

Thank You

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